tokens (words or subwords). This helps maintain conciseness and potentially reduces processing costs associated with longer responses.

NOTE For this book's examples and projects, 150 tokens will generally be sufficient for the initial conversational aspects. However, as you progress to more complex features (such as detailed content generation, summarization of longer texts, or multiturn conversations requiring deeper context), you will likely need to adjust this value upward.

For simplicity, this initial implementation focuses on extracting core information from the OpenAI API response. It constructs a message object containing the response ID, creation time, a designated role (assistant), and the actual content. However, in production environments, additional processing and verification steps would be equally important. These could involve

- *Error handling and retries*—Implementing robust error handling to gracefully handle potential failures during communication with the OpenAI API. This should include mechanisms for retrying failed requests. We will explore comprehensive error-handling strategies in chapter 8.
- *Response validation*—Incorporating logic to verify the validity of the generated responses. This should involve checks for relevance, coherence, or alignment with predefined safety guidelines.

2.5.6 Tests

The tests folder houses a basic testing framework that combines unit tests for individual components. While this doesn't provide exhaustive coverage, establishing a testing infrastructure with tests encompassing various use cases (both normal and unexpected) is essential.

To execute the test suite, navigate to the project's root directory and run the following command:

```
$ npm run test:unit -w ch02/chat-client
```

This command should execute the tests and produce a report detailing the results. The following is the output when running those tests:

```
$ npm run test:unit -w ch02/chat-client
✓ tests/unit/ChatPage.test.jsx (3)
   ✓ ChatPage component (3)
     ✓ renders initial hello message
     ✓ updates input value while typing
     ✓ displays loading state while fetching response

Test Files  1 passed (1)
      Tests  3 passed (3)
   Start at  16:14:32
   Duration  232ms
```